

Name: _____ Date: _____

Period: _____

1. For each rational function below, find all holes and vertical asymptotes. Show your factoring and multiplicity comparison.

(a) $r(x) = \frac{x^2 - 5x + 6}{x - 3}$

Factor: _____

Holes: _____ VAs: _____

(b) $r(x) = \frac{x^2 - x - 12}{(x - 4)(x + 1)}$

Factor: _____

Holes: _____ VAs: _____

(c) $r(x) = \frac{x^2 + 3x}{x^2 + 3x + 2}$

Factor: _____

Holes: _____ VAs: _____

2. **Explain:** In your own words, explain the difference between a hole and a vertical asymptote in terms of the multiplicities m (numerator) and n (denominator) of a shared zero.

3. A rational function $r(x)$ has a hole at $x = 3$ with $\lim_{x \rightarrow 3} r(x) = 5$.

(a) Write the limit statement using proper notation. _____

(b) Explain in a complete sentence what this limit statement means in the context of the graph of r . _____

(c) Is $r(3)$ defined? Circle: **Yes** / **No** Explain briefly: _____

4. Consider $r(x) = \frac{x^3 - x^2}{x^2 - x}$.

(a) Factor the numerator and denominator completely.

Numerator: _____ Denominator: _____

(b) Identify all shared zeros and compare multiplicities to classify each as hole or VA.

Zero	m	n	Hole or VA?
_____	_____	_____	_____
_____	_____	_____	_____

(c) Find the simplified form and the coordinates of all holes.

Simplified: _____ Holes: _____

5. **(AP-Style Multiple Choice)** For the function $r(x) = \frac{x^2 - 4}{x^2 - x - 2}$, which of the following correctly identifies all holes and vertical asymptotes?

- (A) Hole at $x = 2$; VA at $x = -1$
- (B) Hole at $x = -2$; VA at $x = 1$
- (C) VA at $x = 2$ and $x = -1$; no holes
- (D) Hole at $x = 2$; VA at $x = -1$; VA at $x = -2$

Answer: _____ Show your work:
